

Nikolai Slavov

CONTACT INFORMATION

Mugar # 334, Boston, MA 02115
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WWW: northeastern.edu/slavovlab
ORCID ID: [0000-0003-2035-1820](https://orcid.org/0000-0003-2035-1820)

EDUCATION

Princeton University, Princeton, USA
Ph.D. in Molecular and Quantitative Cell Biology **06.2010**
M.A. in Molecular Biology **02.2006**
Dissertation: Universality, specificity and regulation of *S. cerevisiae* growth rate response in different carbon sources and nutrient limitations
Mentor: David Botstein
Massachusetts Institute of Technology (MIT), Cambridge, USA
B.S. in Biology **06.2004**

ACADEMIC APPOINTMENTS

Northeastern University, Boston, USA
Department of Bioengineering
Assistant Professor **2015–present**
Broad Institute of Harvard and MIT, Cambridge, USA
Proteomics Platform **2015–2017**
Harvard University, Cambridge, USA
Departments of Systems Biology and Statistics **2014–2015**
Massachusetts Institute of Technology (MIT), Cambridge, USA
Departments of Biology and Physics
Postdoc in the van Oudenaarden Laboratory **2011–2013**

AWARDS & FELLOWSHIPS

✓ NIH Director's New Innovator Award (\$ 1,500,000 + 852,750) **2016**
✓ NEU Tier I Award (\$ 50,000) **2016**
✓ Broad Institute of Harvard and MIT SPARC Award (\$ 100,000) **2014**
✓ Princeton University Dean's Award **2010**
✓ IRCSET Postgraduate Research Fellowship (72,000 €) **2007**
✓ Finalist in the Young European Entrepreneur Competition **2006**
✓ Princeton Graduate Fellowship (\$ 45,000) **2004**
✓ MIT Undergraduate Fellowship (\$ 125,000) **2001**
✓ Eureka Fellowship for Academic Excellence (10,000 €) **2000**
✓ Bronze Medal in the 31st International Chemistry Olympiad (*IChO*) **1999**
✓ National Diploma for Exceptional Achievements in Chemistry **1999**

2018

2018

- ✓ Emmott EP, Jovanovic M & **Slavov N.**✉
Ribosome stoichiometry: from form to function
Trends in Biochemical Sciences, DOI: [10.1016/j.tibs.2018.10.009](#) [OA]
- ✓ Malioutov D., Chen T., Jaffe J., Airoidi E., Budnik B., **Slavov N.**✉
Quantifying homologous proteins and proteoforms
Molecular & Cellular Proteomics, DOI: [10.1074/mcp.TIR118.000947](#) [OA]
- ✓ Budnik B., Levy E., Harmange G. & **Slavov N.**✉
SCoPE-MS: mass-spectrometry of single mammalian cells quantifies proteome heterogeneity during cell differentiation
Genome Biology, DOI: [10.1186/s13059-018-1547-5](#) [OA]
↔ [SCoPE-MS – We can finally do single cell proteomics!!!](#), highlight by *Proteomics News*
↔ [Researchers Apply Mass Spec to Single-Cell Proteomics](#), highlight by *GenomeWeb*
↔ [Interview](#), highlight by *Front Line Genomics*
- ✓ Levy E. & **Slavov N.**✉
Single cell protein analysis for systems biology
Essays in Biochemistry, DOI: [10.1042/EBC20180014](#)
- ✓ Specht H. & **Slavov N.**✉
Transformative opportunities for single cell proteomics
Journal of Proteome Research, 17 (8), 2565 - 2571 , DOI: [10.1021/acs.jproteome.8b00257](#)
↔ Editorial highlight: [Innovations in Proteomics: The Drive to Single Cells](#)

2017

2017

- ✓ Franks A., Airoidi E.M., **Slavov N.**✉
Post-transcriptional regulation across human tissues
PLoS Computational Biology, 13(5): e1005535, DOI: [110.1371/journal.pcbi.1005535](#) [OA]
↔ [How Statistics Weakened mRNAs Predictive Power](#) highlight by *The Scientist*
- ✓ Saleh D, Najjar M, Zelic M, Shah S, Nogusa S, Polykratis A, Paczosa MK, Gough PJ, Bertin J, Whalen M, Fitzgerald KA, **Slavov N.**, Pasparakis M, Balachandran S, Kelliher M, Meccas J, Degterev A.
Kinase Activities of RIPK1 and RIPK3 Can Direct IFN- Synthesis Induced by Lipopolysaccharide.
The Journal of Immunology, 198(11):4435-4447, DOI: [10.4049/jimmunol.1601717](#)

2016

2016

- ✓ Klionsky, *et al.*, **Slavov N.**, *et al.*
Guidelines for the use and interpretation of assays for monitoring autophagy
Autophagy, vol. 12, issue 1, 1–222, DOI: [10.1080/15548627.2015.1100356](#)
- ✓ Di Luca A, Hamill RM, Mullen AM, **Slavov N.**, Elia G.
Comparative Proteomic Profiling of Divergent Phenotypes for Water Holding Capacity across the Post Mortem Ageing Period in Porcine Muscle Exudate
PLoS One, 11(3): e0150605, DOI: [10.1371/journal.pone.0150605](#) [OA]

2015

2015

- ✓ **Slavov N.** 
 Making the most of peer review
eLife, 4:e12708, DOI: [10.7554/eLife.12708](https://doi.org/10.7554/eLife.12708) [\[OA\]](#)
- ✓ **Slavov N.** , Semrau S., Airoidi E.M., Budnik B., van Oudenaarden A.
 Differential stoichiometry among core ribosomal proteins
Cell Reports, vol. 13, issue 5, 865–873, DOI: [10.1016/j.celrep.2015.09.056](https://doi.org/10.1016/j.celrep.2015.09.056) [\[OA\]](#)
 ⇨ [All Ribosomes Are Created Equal. Really?](#), highlight by *Trends in Biochemical Sciences*
 ⇨ [New direction for tissue engineering and cancer therapeutics](#), highlight by *Science X*
- ✓ Alvarez JR, Zhiqiang B, Xu D, Yuan B, Lo KA, Yoon MJ, Lim YC, Knoll M, **Slavov N.**, *et al.*
 De-Novo Reconstruction of Adipose Tissue Transcriptomes Reveals Long Non-coding RNA Regulators of Brown Adipocyte Development
Cell Metabolism, vol. 21, issue 5, 764–776, DOI: [10.1016/j.cmet.2015.04.003](https://doi.org/10.1016/j.cmet.2015.04.003)

Started Slavov Lab at Northeastern University, August 2015

2014

2014

- ✓ Malioutov D. , **Slavov N.** 
 Convex Total Least Squares
Journal of Machine Learning Research, W&CP, vol. 32, issue 1, 109–117 [\[OA\]](#)
- ✓ **Slavov N.** , Budnik B., Schwab D., Airoidi E., van Oudenaarden A. 
 Constant Growth Rate Can Be Supported by Decreasing Energy Flux and Increasing Aerobic Glycolysis
Cell Reports vol. 7, issue 3, 705–714 [\[OA\]](#)

2013

2013

- ✓ **Slavov N.** , Carey, J., Linse, S. 
 Calmodulin transduces Ca^{+2} oscillations into differential regulation of its target proteins
ACS Chemical Neuroscience, vol. 4, 601–612 [\[OA\]](#)
- ✓ **Slavov N.** , Botstein D. 
 Decoupling Nutrient Signaling from Growth Rate Causes Aerobic Glycolysis and Dereglulation of Cell Size and Gene Expression
Molecular Biology of the Cell, vol. 24, issue 2, 157–168 [\[OA\]](#)

2012

2012

- ✓ **Slavov N.** , van Oudenaarden A. 
 How to Regulate a Gene: To Repress or to Activate?
Molecular Cell, vol. 46, issue 5, 551–552 [\[OA\]](#)

- ✓ **Slavov N.**[✉], Airoidi E.M., van Oudenaarden A., Botstein D.[✉]
A Conserved Cell Growth Cycle Can Account for the Environmental Stress Responses of Divergent Eukaryotes
Molecular Biology of the Cell, vol. 23, no. 10, 1986–1997 [\[OA\]](#)

2011

2011

- ✓ **Slavov N.**[✉], Macinskas J., Caudy A., Botstein D.[✉]
Metabolic Cycling without Cell Division Cycling in Respiring Yeast
PNAS, vol. 108, no. 47, 19090–19095 [\[OA\]](#)
- ✓ **Slavov N.**[✉], Botstein D.[✉]
Coupling among Growth Rate Response, Metabolic Cycle and Cell Division Cycle in Yeast
Molecular Biology of the Cell, vol. 22, no. 12, 1997–2009 [\[OA\]](#)

2010

2010

- ✓ **Slavov N.**[✉]
Inference of Sparse networks with Unobserved Variables. Application to Gene Regulatory Networks
Journal of Machine Learning Research, W&CP, vol. 9, 757–764 [\[OA\]](#)
- ✓ **Slavov N.**[✉], Botstein, D.
Universality, Specificity and Regulation of *S. cerevisiae* Growth Rate Response in Different Carbon Sources and Nutrient Limitations
Dissertation, Princeton University [\[OA\]](#)
- ✓ Silverman SJ., **Slavov N.**, Petti A., Parsons L., Briehof R., Thiberge S., Zenklusen D., Gandhi SJ., Larson D., Singer R., Botstein D.[✉]
Metabolic cycling in single yeast cells from unsynchronized steady-state populations limited on glucose or phosphate
PNAS, vol. 107, no. 15, 6946–6951 [\[OA\]](#)

2009 and before

2009 and before

- ✓ **Slavov N.**[✉], Dawson KA. (2009)
Correlation Signature of the Macroscopic States of the Gene Regulatory Network in Cancer
PNAS, vol. 106, no. 11, 4079–4084 (“In This Issue” article) [\[OA\]](#)
- ✓ Tagkopoulos I.[✉], **Slavov N.**[✉], Kung S.[✉] (2005)
Multi-Class Biclustering and Classification Based on Modeling of Gene Regulatory Networks
5th IEEE Symposium on Bioinformatics and Bioengineering (BIBE’05).

- ▷ Specht H, Harmange G, Perlman DH, Emmott E, Niziolek Z, Budnik B, **Slavov N.** (2018)
Automated sample preparation for high-throughput single-cell proteomics
bioRxiv, DOI: [10.1101/399774](#) [OA]
- ▷ Chen A, Franks A, **Slavov N.** (2018)
DART-ID increases single-cell proteome coverage
bioRxiv, DOI: [10.1101/399121](#) [OA]
- ▷ Levy E. & **Slavov N.** (2018)
Single cell protein analysis for systems biology
PeerJ Preprint, DOI: [10.7287/peerj.preprints.26965v1](#) [OA]
- ▷ Emmott EP, Jovanovic M & **Slavov N.** (2018)
Ribosome stoichiometry: from form to function
PeerJ Preprint, DOI: [10.7287/peerj.preprints.26991v1](#) [OA]
- ▷ Specht H. & **Slavov N.** (2018)
Routinely quantifying single cell proteomes: A new age in quantitative biology and medicine
PeerJ Preprint, DOI: [10.7287/peerj.preprints.26821v1](#) [OA]
- ▷ Malioutov D., Chen T., Jaffe J., Airoidi E., Carr S., Budnik B., **Slavov N.** (2017)
Quantifying homologous proteins and proteoforms
bioRxiv, DOI: [10.1101/168765](#) [OA]
- ▷ Berg P., Budnik B., **Slavov N.**, Semrau S. (2017)
Dynamic post-transcriptional regulation during embryonic stem cell differentiation
bioRxiv, DOI: [10.1101/123497](#) [OA]
- ▷ Budnik B., Levy E., **Slavov N.** (2017)
Mass-spectrometry of single mammalian cells quantifies proteome heterogeneity during cell differentiation
bioRxiv, DOI: [10.1101/102681](#) [OA]
 - ↪ [SCoPE-MS – We can finally do single cell proteomics!!!](#), highlight by *Proteomics News*
 - ↪ [Researchers Apply Mass Spec to Single-Cell Proteomics](#), highlight by *GenomeWeb*
 - ↪ [Interview](#), highlight by *Front Line Genomics*
- ▷ **Slavov N.** (2015)
From differential transcription of ribosomal proteins to differential structure of ribosomes
PeerJ Preprint, DOI: [10.7287/peerj.preprints.1504v1](#) [OA]
- ▷ Franks A., Airoidi E.M., **Slavov N.** (2015)
Post-transcriptional regulation across human tissues
bioRxiv, DOI: [10.1101/020206](#) [OA]
- ▷ **Slavov N.**, Botstein, D., Caudy A. (2014)
Extensive regulation of metabolism and growth during the cell division cycle
bioRxiv, DOI: [10.1101/005629](#) [OA]

- ▷ **Slavov N.** , Semrau S., Airoidi E.M., Budnik B., van Oudenaarden A. (2014)
Differential stoichiometry among core ribosomal proteins
bioRxiv, DOI: [10.1101/005553](https://doi.org/10.1101/005553) **[OA]**

OPINIONS &
POPULAR SCIENCE
PUBLICATIONS

- ▷ **Slavov N.**  (2018) We Must Demand Evidence of Peer Review
The Scientist, [Link](#)
- ▷ **Slavov N.**  (2017) What was rate limiting was the idea, not the technology
Front Line Genomics, [Link](#)
- ▷ **Slavov N.**  (2015) Making the most of peer review
eLife, [Link](#)
- ▷ **Slavov N.**  (2014) Accomplishments Over Accolades!
The Scientist, [Link](#)
- ▷ **Slavov N.**  (2014) Discoveries lie hidden behind the façade of popular assumptions
Cell Reporter, [Link](#)
- ▷ **Slavov N.**  (2013) The Best Projects Are Least Obvious
The Scientist, [Link](#)
- ▷ **Slavov N.**  (2013) A Wonderful Christmas with Mysterious Oscillations
PubChase, [Link](#)
- ▷ **Slavov N.**  (2012) The Mission of MIT
The MIT Tech, [Link](#)

ORGANIZED
CONFERENCES

- [Single Cell Proteomics](#), Northeastern University **June 10–12 2019**
- [Chair for the Single Cell Proteomics session](#), HUPO **Oct 1–3 2018**
- [Single Cell Proteomics](#), Northeastern University **June 9–10 2018**

SELECTED
RESEARCH TALKS

- Transformative opportunities for single cell proteomics
The Greater Boston Mass Spectrometry Discussion Group **Dec.13.2018**
- Understanding and controlling cell differentiation based on comprehensive mass-spec quantification of proteins and signaling in single cells
UPENN Single Cell Biology Symposium, [Video](#) **Nov.13.2018**

- Chair for the Single Cell Proteomics session of HUPO
Automated sample preparation for high-throughput single-cell proteomics **Oct.3.2018**
- Using single-cell proteomics to engineer directed cell differentiation
Bioengineering & Translational Medicine Conference **Sept.27.2018**
- Analyzing single cell proteomics data
ETH, Biognosys Discovery Proteomics Seminar **Jul.7.2018**
- Progress, challenges and standards for single cell proteomics
Single-cell proteomics conference, [Video](#) **June.9.2018**
- Understanding and controlling cell differentiation using single-cell mass-spectrometry
HMS Theory Lunch **May.18.2018**
- Mass-spectrometry of single mammalian cells quantifies proteome heterogeneity during cell differentiation
CSHL Meeting: Single Cell Analyses **Nov.10.2017**
- Single cell proteomics quantifies proteome heterogeneity during cell differentiation
Annual Single Cell Analysis USA Congress, Boston **Oct.23.2017**
- Quantifying protein heterogeneity during the differentiation of single stem cells
Global CESI-MS Symposium **Oct.06.2017**
- Single cell proteomics quantifies proteome heterogeneity during cell differentiation
Festival of Genomics, Boston **Oct.04.2017**
- Exploring and characterizing system-level proteome heterogeneity during cell differentiation by single cell mass spectrometry
HUPO, Dublin **Sept.18.2017**
- From differential transcription of ribosomal proteins to differential structure of ribosomes
Genetics, Genomics and Beyond, Calico **Sept.16.2017**
- Mass-spectrometry of single mammalian cells quantifies proteome heterogeneity during cell differentiation
American Society for Mass Spectrometry Meeting **June.05.2017**
- Principles of ribosome-mediated translational regulation
The Center for Engineering in Medicine, Massachusetts General Hospital **Mar.17.2017**
- Principles of ribosome-mediated translational regulation
Systems Biology Seminar, Boston University **Oct.13.2016**
- Differential stoichiometry among core ribosomal proteins
CSHL Translational Control, Cold Spring Harbor Laboratory **Sept.08.2016**

- Ribosome-mediated translational regulation: Direct methods for quantifying the ribosome code
Center for Systems Biology, Columbia University **Apr.20.2016**
- Coordinating metabolism and ribosome-mediated translational regulation
Department of Biology, New York University **Apr.19.2016**
- Principles of ribosome-mediated translational regulation
Center for Interdisciplinary Research on Complex Systems **Apr.12.2016**
- Quantifying homologous proteins and proteoforms
US HUPO Conference **Mar.14.2016**
- Coordination among cell growth, cell division and gene expression
Broad Institute of MIT and Harvard, YSB **Dec.9.2015**
- Coordinating cell growth and metabolism with ribosome-mediated translational regulation
Northeastern University, Biology Department Colloquium **Oct.19.2015**
- Quantifying protein isoforms
Broad Institute of MIT and Harvard **Sep.14.2015**
- Variable stoichiometry among core ribosomal proteins
2014 ASCB/IFCB Annual Meeting **Dec.8.2014**
- Direct quantification of highly homologous proteins by mass-spectrometry indicates physiological changes in the protein composition of the ribosome
Broad Institute of MIT and Harvard, CBBO **Jun.25.2014**
- Variable stoichiometry among core ribosomal proteins
Broad Institute of MIT and Harvard **Jun.19.2014**
- Quantification of translational regulation of homologous proteins with unprecedented accuracy
Harvard HMS Systems Biology Theory Lunch **Mar.21.2014**
- Exponentially growing cells can support a constant growth rate by very different metabolic strategies
Fourth Annual Physical Sciences in Oncology Meeting, NCI **Apr.18.2013**
- Experimentally identified trade-offs of aerobic glycolysis
Center of Cancer Systems Biology at Tufts **Dec.18.2012**
- Trade-offs of aerobic glycolysis
Harvard HMS Systems Biology Theory Lunch **Sep.12.2012**
- The dynamics of exponential growth
GSA Yeast Meeting, Princeton University **Aug.5.2012**
- The dynamics of exponential cell growth at constant growth rate
MIT Physical Sciences-Oncology Center **Jun.11.2012**

- The dynamics of exponential growth
Broad Institute of Harvard and MIT **Apr.17.2012**
- Metabolic cycling without cell division cycling
Bauer Forum at Harvard FAS Systems Biology **Apr.15.2011**
- Inference of Sparse Networks with Unobserved Variables. Application to Gene Regulatory Networks
International Conference on Artificial Intelligence and Statistics **May.14.2010**
- Dynamics in the calcium-calmodulin signaling network
International Conference on Chaos and Nonlinear Dynamics **Jan.8.2006**

PROFESSIONAL ACTIVITIES

- **Peer-review:** PNAS, Nature Molecular and Structural Biology, Molecular Biology of the Cell, PLoS, Nucleic Acids Research, Bioinformatics, Annals of Applied Statistics, Cell Reports, eLife, Nature Communications || [Publons Profile](#)
- **Editorial roles:** PeerJ, bioRxiv Affiliate

TEACHING & RESIDENTIAL ADVISING

- | | |
|---|--------------------|
| Mathematical Methods for Engineers , Northeastern University | 2016 – 2018 |
| Introduction to major concepts and methods in linear algebra differential equations | |
| Method and Logic in Quantitative Biology , Princeton University | 2009 – 2010 |
| Introduction to major ideas and concepts in quantitative biology based on primary literature | |
| Residential Graduate Student at Forbes College , Princeton University | 2009 – 2010 |
| Advised students; organized language tables, sport events and trips | |
| Integrated Quantitative Introduction to Natural Sciences , Princeton University | 2006 – 2010 |
| Interdisciplinary curriculum including physics, computer science, biochemistry, genetics, physiology and emphasizing quantitative problem solving and the connections among these disciplines | |
| Terrascope , MIT | 2002 – 2003 |
| Selected for two subsequent years to lead undergraduate student groups working on complex real-world problems | |
| Biochemistry , MIT | 2003 – 2004 |
| Taught precepts and helped undergraduate students prepare for examinations | |